

GENERAL SCIENCE: PHYSICS FUNDAMENTALS

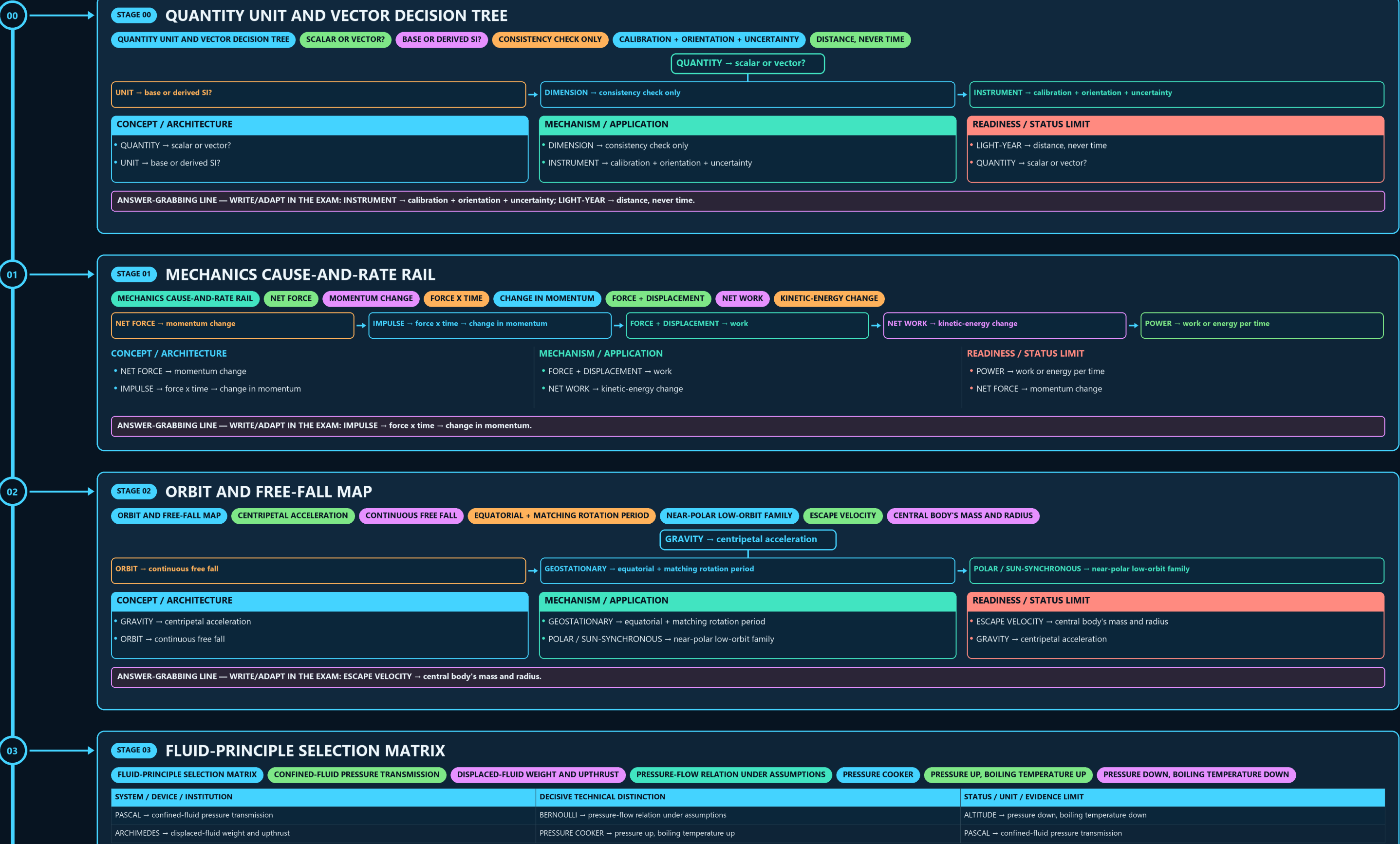
QUANTITY UNIT AND VECTOR → MECHANICS CAUSE-AND-RATE RAIL → ORBIT AND FREE-FALL MAP → FLUID-PRINCIPLE SELECTION MATRIX → THERMAL STATE AND TRANSFER → WAVE SOUND AND SPECTRUM

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g2 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



CONCEPT / ARCHITECTURE

- GRAVITY → centripetal acceleration
- ORBIT → continuous free fall

MECHANISM / APPLICATION

- GEOSTATIONARY → equatorial + matching rotation period
- POLAR / SUN-SYNCHRONOUS → near-polar low-orbit family

READINESS / STATUS LIMIT

- ESCAPE VELOCITY → central body's mass and radius
- GRAVITY → centripetal acceleration

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ESCAPE VELOCITY → central body's mass and radius.

03

STAGE 03

FLUID-PRINCIPLE SELECTION MATRIX

FLUID-PRINCIPLE SELECTION MATRIX

CONFINED-FLUID PRESSURE TRANSMISSION

DISPLACED-FLUID WEIGHT AND UPTHrust

PRESSURE-FLOW RELATION UNDER ASSUMPTIONS

PRESSURE COOKER

PRESSURE UP, BOILING TEMPERATURE UP

PRESSURE DOWN, BOILING TEMPERATURE DOWN

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
PASCAL → confined-fluid pressure transmission	BERNOULLI → pressure-flow relation under assumptions	ALTITUDE → pressure down, boiling temperature down
ARCHIMEDES → displaced-fluid weight and upthrust	PRESSURE COOKER → pressure up, boiling temperature up	PASCAL → confined-fluid pressure transmission

SYSTEM / DEVICE / INSTITUTION

- PASCAL → confined-fluid pressure transmission
- ARCHIMEDES → displaced-fluid weight and upthrust

DECISIVE TECHNICAL DISTINCTION

- BERNOULLI → pressure-flow relation under assumptions
- PRESSURE COOKER → pressure up, boiling temperature up

STATUS / UNIT / EVIDENCE LIMIT

- ALTITUDE → pressure down, boiling temperature down
- PASCAL → confined-fluid pressure transmission

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PRESSURE COOKER → pressure up, boiling temperature up.

04

STAGE 04

THERMAL STATE AND TRANSFER LADDER

THERMAL STATE AND TRANSFER LADDER

THERMAL STATE

TRANSFER DUE TO TEMPERATURE DIFFERENCE

SPECIFIC HEAT

TEMPERATURE-CHANGE REQUIREMENT

LATENT HEAT

PHASE CHANGE WITHOUT TEMPERATURE CHANGE

DISTINCT ROUTES

TEMPERATURE → thermal state

HEAT → transfer due to temperature difference

SPECIFIC HEAT → temperature-change requirement

LATENT HEAT → phase change without temperature change

STARTING CONDITION / INPUT

- TEMPERATURE → thermal state
- HEAT → transfer due to temperature difference

SCIENTIFIC OR ENGINEERING MECHANISM

- SPECIFIC HEAT → temperature-change requirement
- LATENT HEAT → phase change without temperature change

OUTPUT / FAILURE BOUNDARY

- RADIATION → distinct routes
- TEMPERATURE → thermal state

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CONDUCTION | CONVECTION | RADIATION → distinct routes.

05

STAGE 05

WAVE SOUND AND SPECTRUM BANDS

WAVE SOUND AND SPECTRUM BANDS

FREQUENCY X WAVELENGTH

MEDIUM REQUIRED

FREQUENCY CLASSES

DIFFERENT ELECTROMAGNETIC BANDS

WAVE → speed = frequency x wavelength

SOUND → medium required

ULTRASOUND → frequency classes

RADIO → MICROWAVE → IR → VISIBLE → UV → X-RAY → GAMMA

RADAR / VLC → different electromagnetic bands

CONCEPT / ARCHITECTURE

- WAVE → speed = frequency x wavelength
- SOUND → medium required

MECHANISM / APPLICATION

- ULTRASOUND → frequency classes
- RADIO → MICROWAVE → IR → VISIBLE → UV → X-RAY → GAMMA

READINESS / STATUS LIMIT

- RADAR / VLC → different electromagnetic bands
- WAVE → speed = frequency x wavelength

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RADIO → MICROWAVE → IR → VISIBLE → UV → X-RAY → GAMMA.

06

STAGE 06

MIRROR LENS AND FIBRE RAY LOGIC

MIRROR LENS AND FIBRE RAY LOGIC

CONVEX LENS

CONCAVE LENS

DENSER TO RARER + CRITICAL ANGLE

TOTAL INTERNAL REFLECTION

MIRROR → reflection

LENS → refraction

CONVEX LENS → converging

CONCAVE LENS → diverging

CONCEPT / ARCHITECTURE

MECHANISM / APPLICATION

READINESS / STATUS LIMIT

GENERAL SCIENCE: PHYSICS FUNDAMENTALS • TILE 2/4 • same master rows 2387-5621 of 10396 • continues below

06

CONCEPT / ARCHITECTURE

- WAVE → speed = frequency x wavelength
- SOUND → medium required

MECHANISM / APPLICATION

- ULTRASOUND → frequency classes
- RADIO → MICROWAVE → IR → VISIBLE → UV → X-RAY → GAMMA

READINESS / STATUS LIMIT

- RADAR / VLC → different electromagnetic bands
- WAVE → speed = frequency x wavelength

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RADIO → MICROWAVE → IR → VISIBLE → UV → X-RAY → GAMMA.

06

STAGE 06

MIRROR LENS AND FIBRE RAY LOGIC

MIRROR LENS AND FIBRE RAY LOGIC

CONVEX LENS

CONCAVE LENS

DENSER TO RARER + CRITICAL ANGLE

TOTAL INTERNAL REFLECTION

MIRROR → reflection

LENS → refraction

CONVEX LENS → converging

CONCAVE LENS → diverging

CONCEPT / ARCHITECTURE

- MIRROR → reflection
- LENS → refraction

MECHANISM / APPLICATION

- CONVEX LENS → converging
- CONCAVE LENS → diverging

READINESS / STATUS LIMIT

- DENSER TO RARER + CRITICAL ANGLE → total internal reflection
- MIRROR → reflection

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DENSER TO RARER + CRITICAL ANGLE → total internal reflection.

07

STAGE 07

CIRCUIT TOPOLOGY AND SAFETY BOARD

CIRCUIT TOPOLOGY AND SAFETY BOARD

WORK PER CHARGE

CHARGE PER TIME

OPPOSITION TO CURRENT

COMMON CURRENT

COMMON VOLTAGE

SHOCK PROTECTION

CONCEPT / ARCHITECTURE

- VOLTAGE → work per charge
- CURRENT → charge per time
- RESISTANCE → opposition to current

MECHANISM / APPLICATION

- SERIES → common current
- PARALLEL → common voltage
- FUSE / MCB → overload

READINESS / STATUS LIMIT

- EARTHING → shock protection

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FUSE / MCB → overload | EARTHING → shock protection.

08

STAGE 08

ELECTROMAGNETIC CONVERSION LOOP

ELECTROMAGNETIC CONVERSION LOOP

CHANGING FLUX

INDUCED EMF

MECHANICAL TO ELECTRICAL

AC VOLTAGE CONVERSION

ELECTRICAL TO MECHANICAL

STEADY DC

NO ORDINARY TRANSFORMER ACTION

CONCEPT / ARCHITECTURE

- CHANGING FLUX → induced emf
- GENERATOR → mechanical to electrical

MECHANISM / APPLICATION

- TRANSFORMER → AC voltage conversion
- MOTOR → electrical to mechanical

READINESS / STATUS LIMIT

- STEADY DC → no ordinary transformer action
- CHANGING FLUX → induced emf

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MOTOR → electrical to mechanical; STEADY DC → no ordinary transformer action.

09

STAGE 09

QUANTUM AND NUCLEAR SPLIT PANEL

QUANTUM AND NUCLEAR SPLIT PANEL

THRESHOLD FREQUENCY

INTEGER-SPIN BOSONS MAY SHARE STATE

DIFFERENT MATTER/RADIATION CLASSES

NUCLEAR DECAY CLOCK

MUST REMEMBER

PHYSICS FUNDAMENTALS REQUIRE QUANTITIES, SI UNITS, DIMENSIONS, FRAMES AND

PHOTOELECTRIC → threshold frequency

BOSE-EINSTEIN → integer-spin bosons may share state

GAMMA → different matter/radiation classes

HALF-LIFE → nuclear decay clock

FISSION != FUSION

CONCEPT / ARCHITECTURE

- PHOTOELECTRIC → threshold frequency
- BOSE-EINSTEIN → integer-spin bosons may share state

MECHANISM / APPLICATION

- GAMMA → different matter/radiation classes
- HALF-LIFE → nuclear decay clock

READINESS / STATUS LIMIT

- FISSION != FUSION
- MUST REMEMBER: Physics fundamentals require quantities, SI units, dimensions, frames and...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Physics fundamentals require quantities, SI units, dimensions, frames and.

10

STAGE 10

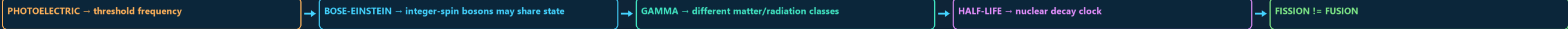
RELATIVITY OBSERVATION EVIDENCE CHAIN

GENERAL SCIENCE: PHYSICS FUNDAMENTALS • TILE 3/4 • same master rows 4775-8009 of 10396 • continues below

09

STAGE 09 QUANTUM AND NUCLEAR SPLIT PANEL

QUANTUM AND NUCLEAR SPLIT PANEL THRESHOLD FREQUENCY INTEGER-SPIN BOSONS MAY SHARE STATE DIFFERENT MATTER/RADIATION CLASSES NUCLEAR DECAY CLOCK MUST REMEMBER PHYSICS FUNDAMENTALS REQUIRE QUANTITIES, SI UNITS, DIMENSIONS, FRAMES AND



CONCEPT / ARCHITECTURE

- PHOTOELECTRIC → threshold frequency
- BOSE-EINSTEIN → integer-spin bosons may share state

MECHANISM / APPLICATION

- GAMMA → different matter/radiation classes
- HALF-LIFE → nuclear decay clock

READINESS / STATUS LIMIT

- FISSION != FUSION
- MUST REMEMBER: Physics fundamentals require quantities, SI units, dimensions, frames and...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Physics fundamentals require quantities, SI units, dimensions, frames and.

10

STAGE 10 RELATIVITY OBSERVATION EVIDENCE CHAIN

RELATIVITY OBSERVATION EVIDENCE CHAIN SPACETIME PREDICTION GRAVITATIONAL WAVE INSTRUMENT SIGNAL SOURCE INFERENCE DISTINCT ASTRONOMY LABELS BOUNDED CLAIM, NOT OMNISCIENCE CLOSE DISTINCTION



STARTING CONDITION / INPUT

- RELATIVITY → spacetime prediction
- GRAVITATIONAL WAVE → instrument signal

SCIENTIFIC OR ENGINEERING MECHANISM

- ANALYSIS → source inference
- PULSAR → distinct astronomy labels

OUTPUT / FAILURE BOUNDARY

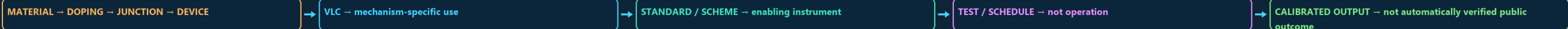
- OBSERVATION → bounded claim, not omniscience
- CLOSE DISTINCTION: Scalar is not vector, speed is not velocity, mass is not weight,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Scalar is not vector, speed is not velocity, mass is not weight.

11

STAGE 11 DEVICE TO VERIFIED-STATUS STAIRCASE

DEVICE TO VERIFIED-STATUS STAIRCASE MECHANISM-SPECIFIC USE ENABLING INSTRUMENT NOT OPERATION CALIBRATED OUTPUT NOT AUTOMATICALLY VERIFIED PUBLIC OUTCOME EVIDENCE LIMIT WRITE THE GOVERNING RELATION, DEFINE



CONCEPT / ARCHITECTURE

- MATERIAL → DOPING → JUNCTION → DEVICE
- VLC → mechanism-specific use

MECHANISM / APPLICATION

- STANDARD / SCHEME → enabling instrument
- TEST / SCHEDULE → not operation

READINESS / STATUS LIMIT

- CALIBRATED OUTPUT → not automatically verified public outcome
- EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Write the governing relation, define...

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Write the governing relation, define...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Write the governing relation, define.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT METROLOGY AND STANDARDS — ESSENTIAL WHEN PRECISION MEASUREMENT IS ELECTRONICS AND DEVICE RESEARCH, LINKING SOLID-STATE PHYSICS TO USABLE INDIA SEMICONDUCTOR MISSION

WHERE BASIC SEMICONDUCTOR PHYSICS BECOMES MANUFACTURING AND DESIGN POLICY. DST NATIONAL QUANTUM MISSION RELEVANT BECAUSE QUANTUM DEVICES, SENSING AND MATERIALS ALL REST DAE/BARC ECOSYSTEM

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH

- CSIR-NPL: metrology and standards — essential when precision measurement is the real bottleneck.
- CSIR-CEERI: electronics and device research, linking solid-state physics to usable components.
- India Semiconductor Mission: where basic semiconductor physics becomes manufacturing and design policy.
- DST National Quantum Mission: relevant because quantum devices, sensing and materials all rest on advanced physics principles.

READINESS, UNIT OR STATUS LIMIT

- DAE/BARC ecosystem: policy-level continuation of nuclear and radiation-related physics in applied form.
- Power electronics, transformers and motors are central to India's electrification and industrial productivity.
- Laser-based tools matter in manufacturing, surveying, surgery and defence optics.
- Semiconductor devices determine telecom, computing, EV power systems, consumer electronics and strategic systems.

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE

- Radiation science supports diagnostics, therapy, sterilization and industrial measurement.
- CAUTION: Limitation 1: a device may work in the lab but fail at scale because thermal management, materials purity or measurement standards are weak.
- CAUTION: Limitation 2: public debate often jumps directly to final products while
- neglecting the physics-heavy middle layers such as metrology, materials and clean-room ecosystems.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.